MORPHOPY: A NEW TOOL FOR IMPACT CRATER MORPHOLOGICAL ANALYSES

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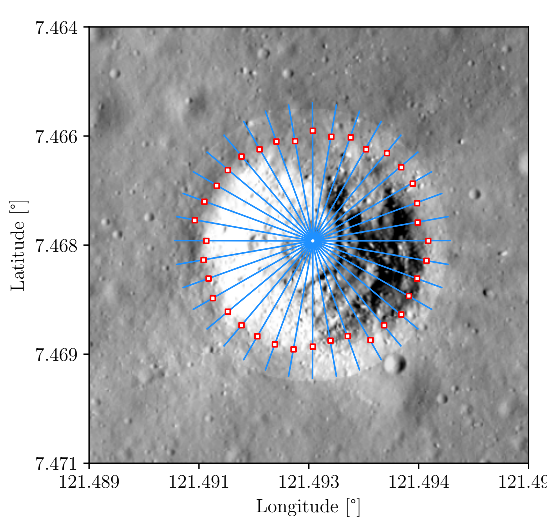
Introduction: Impact craters record the geological history and surface evolution of planetary surfaces[1]. Their depth-to-diameter (d/D) ratio and degradation state encode target surface properties[2], impactor characteristics[3], and terrain age[4]. Nowadays, these features are still measured by hand: subjective, slow, inconsistent[5]. MorphoPy is a Python tool that derives them automatically from a DEM and a Shapefile crater database, including d/D and a new freshness parameter for crater degradation state[6].

Data & Method: MorphoPy needs two inputs: a DTM and a crater identification file giving approximate centre coordinates and diameters obtain from a machine-learning algorithm, manual mapping, or OpenCraterTool[7]. Each crater is clipped to a buffer, the lowest point near its centre is taken as the crater floor, and 36 radial profiles are cast every 10°. The highest point on either side of the floor defines the rim crest. Outputs: a metrics table and a crater shapefile.



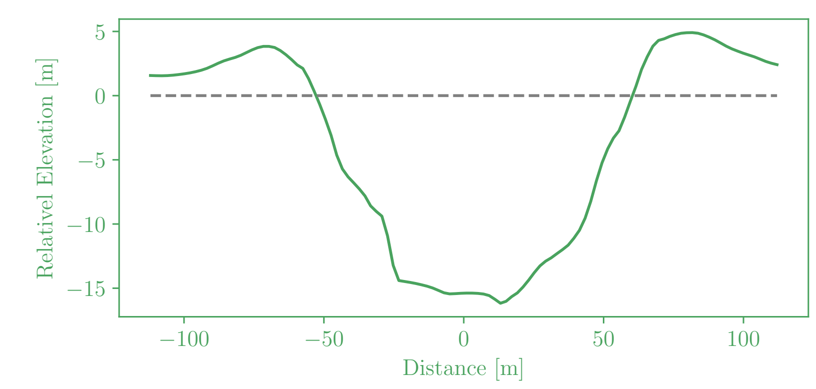
Common metrics: Every metric derives from the same 36 profiles. Diameter is the mean of the 18 rim-to-rim chords; depth the mean drop from the 36 rim points to the crater floor; wall slope the mean profile gradient, following Stopar et al. (2017). Circularity and eccentricity are fitted following Stopper & al., 2017 and Robbins 2018. Each value carries an uncertainty propagated from pixel size and DTM vertical accuracy[8].

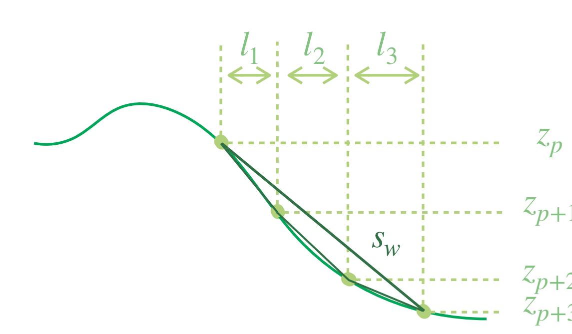
$$D = \frac{1}{18}\sum_{i=0}^{17}\sqrt{\left(x_i - x_{i+18}\right)^2 + \left(y_i - y_{i+18}\right)^2}$$

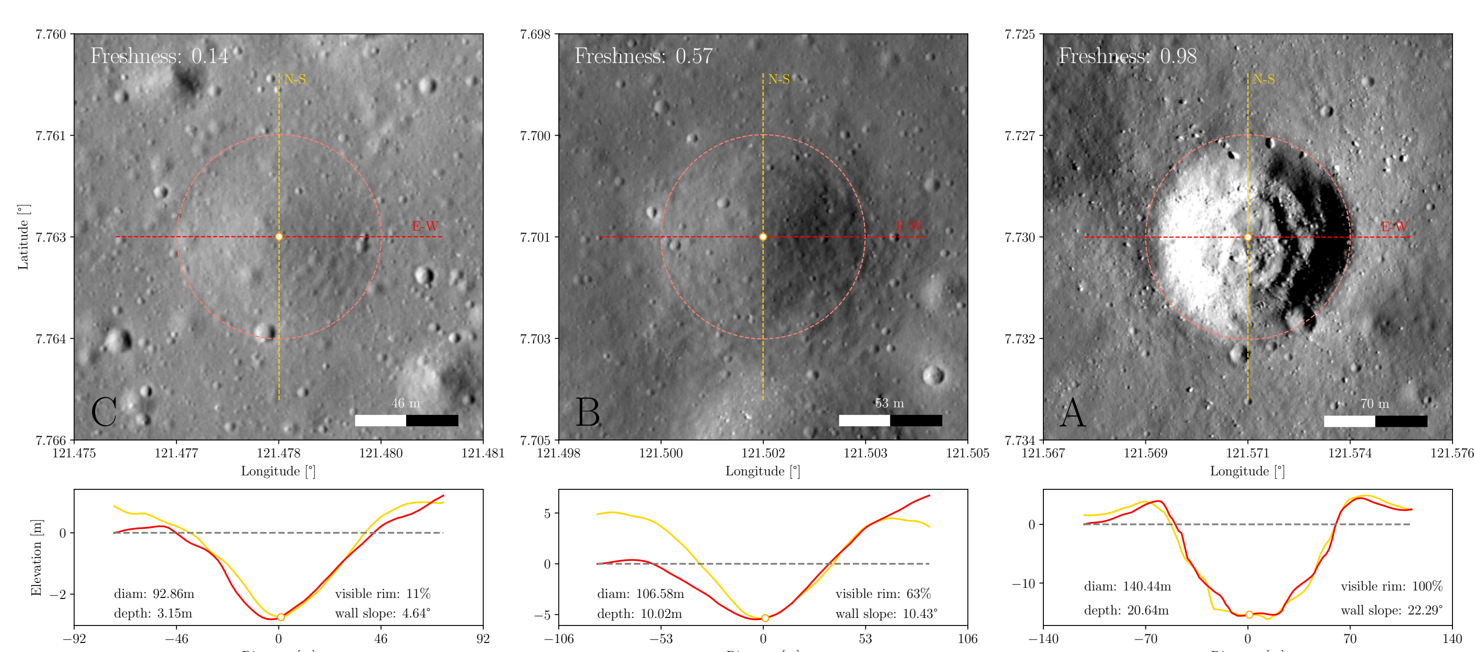
$$d = \frac{1}{36}\sum_{i=0}^{35}\sqrt{z_i - z_{min}}$$

$$s_w = \frac{1}{n}\sum_{i=0}^{n-1} tan^{-1}\left(\frac{z_{p+1} - z_p}{l_i}\right)\cdot\frac{180}{\pi}$$

With: $$l_i = \sqrt{\left(x_{p+1} - x_p\right)^2 + \left(y_{p+1} - y_p\right)^2}$$



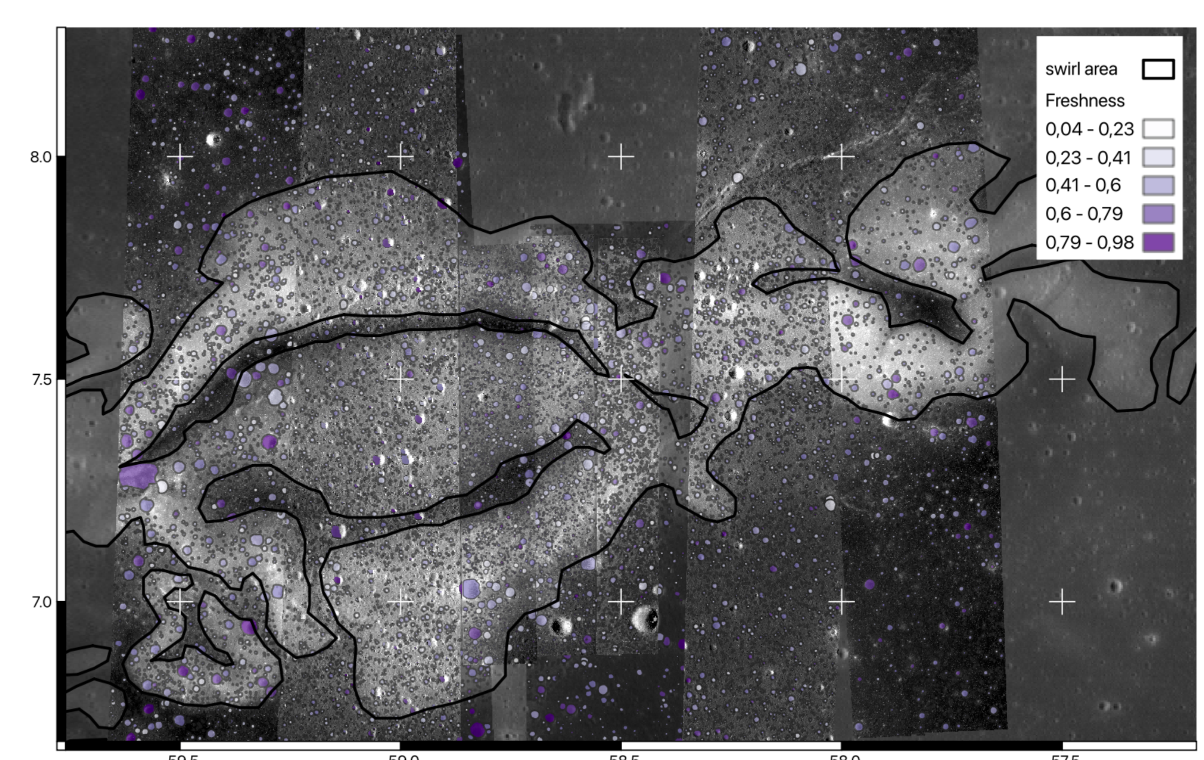




Freshness & d/D: Freshness compresses degradation into a single number between 0 and 1. Two terms: the fraction of half-profiles that still show a rim, and the median wall slope relative to a 24° reference[add]. A crater with a continuous rim scores ≥ 0.66; one with no rim cannot exceed 0.34. The scale maps onto Basilevsky's five classes[9] without visual interpretation. Paired with d/D, it separates preservation state from size.

$$\text{Freshness} = 0.66\left(\frac{\sum_{i=0}^{n}(h_{r(i)} > 0)}{n}\right) + 0.34\left(\frac{m\,s_w}{s_{w_{max}}}\right)$$

Large scale application on the Vertex rover landing site (ReinerGamma): Lunar Vertex will land at Reiner Gamma (7.585° N, 58.725° W) and traverse up to 2 km across the swirl. We ran MorphoPy over 2–5 m LROC NAC DTMs of the region: a YOLOv5 detector proposed ~64,000 craters, of which 22,192 passed the quality filters. The resulting d/D and freshness maps characterise regolith maturity and surface roughness across the landing area before the rover arrives. Across 22,192 craters at Reiner Gamma, depth scales with diameter, but the scatter is governed by preservation state. Fresh craters (freshness 0.8–1) fall between d/D = 0.10 and 0.19; degraded ones (freshness ≤ 0.5) drop below 0.08. This analysis agree with published values for small lunar craters[add ref] — obtained here with an automatic classification.



Conclusion & Perspectives: MorphoPy turns crater morphometry into a reproducible measurement. From one DTM and one crater catalogue, every metric — d/D, wall slope, circularity, freshness — comes out with a propagated uncertainty. Standardised d/D values can be compared across sites and studies, and fed into topographic diffusion models at population scale.

→ Lunar south pole analysis
→ Coupling with AI detection and classification
→ Extension to Mars and Mercury
→ Manuscript under revision workflow release planned for Fall 2026

GEODES

Fonds d'appui à l'engagement étudiant (FAEE) de l'Université de Sherbrooke

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